

Feature Selection Techniques

Aboalhasan Abbas

College of Computer Science & IT — University of Wasit

Most columns in a real dataset do not earn their place.

- Irrelevant: no relationship to the target — a customer's telephone number.
- Redundant: duplicates information already present — age versus date of birth.
- Cost: poor-quality patterns, and a slower mining process.

Feature selection chooses existing columns; it never creates new ones.

- Select: keep a subset of the existing attributes (drop Phone).
- Construct: add a new column from existing ones ($\text{area} = \text{height} \times \text{width}$).
- Extract: build a new representation with new axes (PCA components).
- All three sit under the umbrella of data reduction.

The goal is the smallest subset that still preserves the class distribution.

" The goal of attribute subset selection is to find a minimum set of attributes such that the resulting probability distribution of the data classes is as close as possible to the original distribution obtained using all attributes." — Han, Kamber & Pei, p.104 "

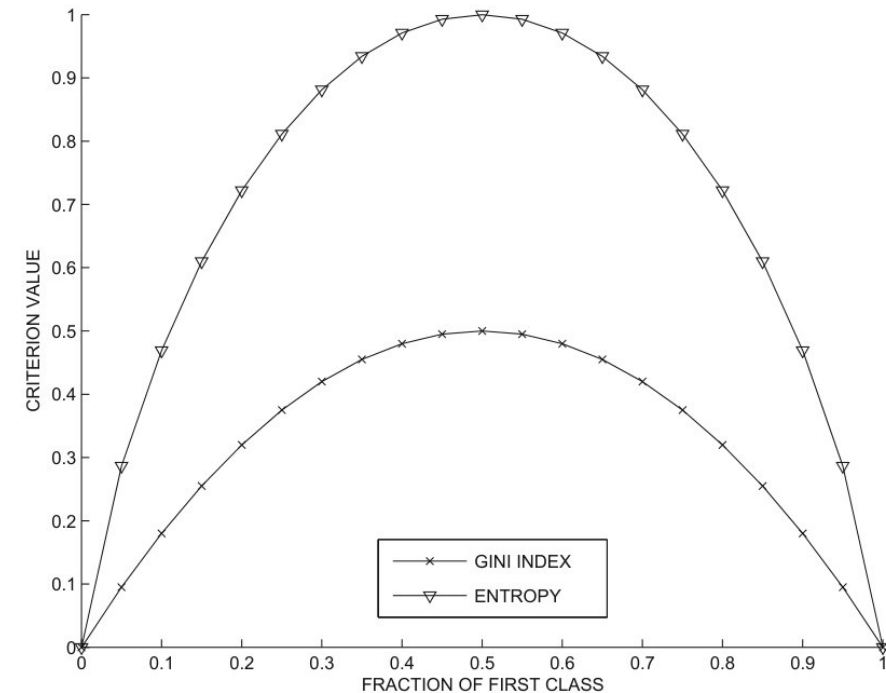
- Minimum set: we are minimising.
- Class distribution preserved: we are constrained.
- Together: a trade-off, not a one-sided goal.

There are 2-to-the-n possible subsets, so exhaustive search is impossible.

- 20 features: 1,048,576 subsets. 30 features: over a billion. 50 features: over a quadrillion.
- H&K: an exhaustive search "can be prohibitively expensive".
- So we search heuristically — greedy: a locally optimal choice, hoping for a near-global optimum.

Both Gini and entropy ask one question: how mixed are the classes?

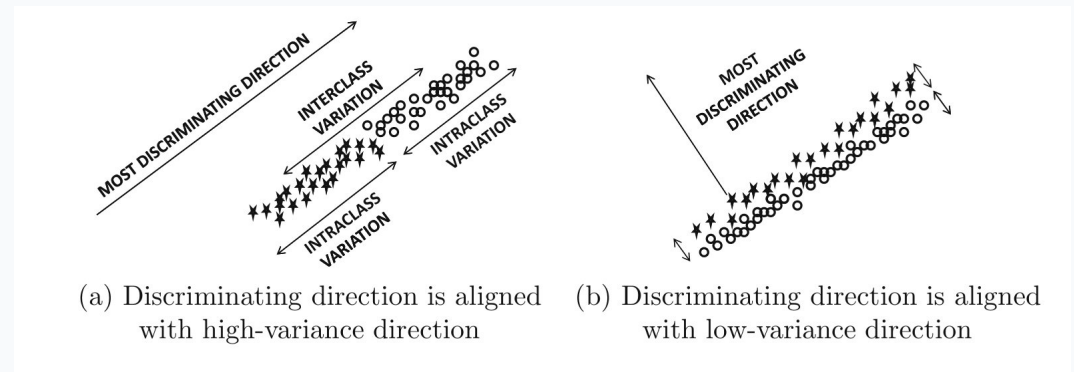
- Gini (Eq. 10.1): 1 minus the sum of squared class fractions.
- Entropy (Eq. 10.3): minus the sum of $p \log_2 p$.
- Both: 0 = perfect separation; higher = more mixing. Lower is better.



Gini and entropy against class-distribution skew (Aggarwal, p.289)

The Fisher score rewards classes that are far apart and tight within.

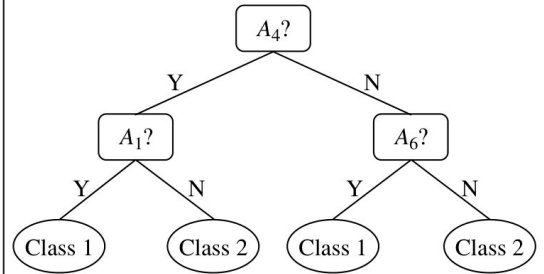
- Eq. 10.5: interclass separation divided by intraclass spread.
- Numerator: how far each class mean sits from the global mean.
- Denominator: how wide each class is. Higher is better.



Fisher's discriminating direction vs class distribution (Aggarwal, p.291)

Four greedy searches all converge on the same reduced subset.

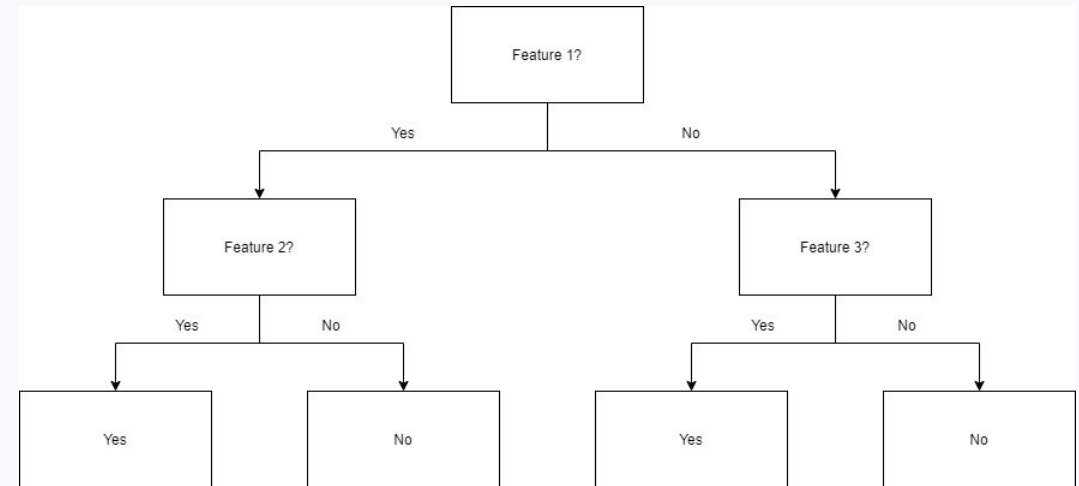
- Forward selection: start empty, add the best each step.
- Backward elimination: start full, remove the worst each step.
- Combination: add the best and remove the worst. Decision tree: keep the attributes it splits on.

Forward selection	Backward elimination	Decision tree induction
Initial attribute set: $\{A_1, A_2, A_3, A_4, A_5, A_6\}$ Initial reduced set: $\{\}$ $\Rightarrow \{A_1\}$ $\Rightarrow \{A_1, A_4\}$ $\Rightarrow$ Reduced attribute set: $\{A_1, A_4, A_6\}$	Initial attribute set: $\{A_1, A_2, A_3, A_4, A_5, A_6\}$ $\Rightarrow \{A_1, A_3, A_4, A_5, A_6\}$ $\Rightarrow \{A_1, A_4, A_5, A_6\}$ $\Rightarrow$ Reduced attribute set: $\{A_1, A_4, A_6\}$	Initial attribute set: $\{A_1, A_2, A_3, A_4, A_5, A_6\}$  <pre>graph TD; A4["A4?"] -- Y --> A1["A1?"]; A4 -- N --> A6["A6?"]; A1 -- Y --> C1_1((Class 1)); A1 -- N --> C2_1((Class 2)); A6 -- Y --> C1_2((Class 1)); A6 -- N --> C2_2((Class 2));</pre> $\Rightarrow$ Reduced attribute set: $\{A_1, A_4, A_6\}$

Greedy methods for attribute subset selection (Han, Kamber & Pei, p.104)

A decision tree keeps only the attributes it actually splits on.

- Each internal node is a test on an attribute.
- Each branch is an outcome; each leaf is a class prediction.
- Attributes that never appear in the tree are assumed irrelevant.

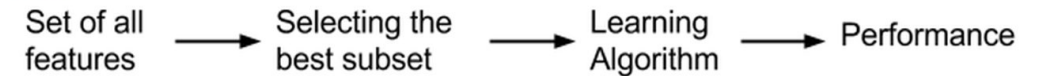


A decision tree: internal tests, branches, and class leaves

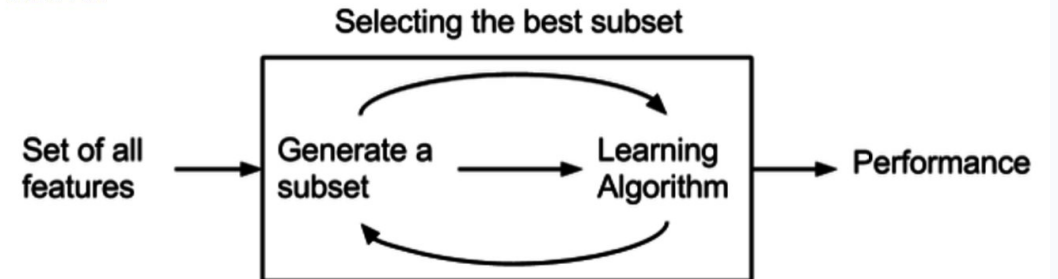
Who computes the score? That single question defines three families.

- Filter: a fixed mathematical criterion — no classifier, cheap, model-agnostic.
- Wrapper: the classifier's own accuracy judges each subset — expensive, model-tied.
- Embedded: the model reveals the features while training — e.g. small weights, Lasso, RFE.

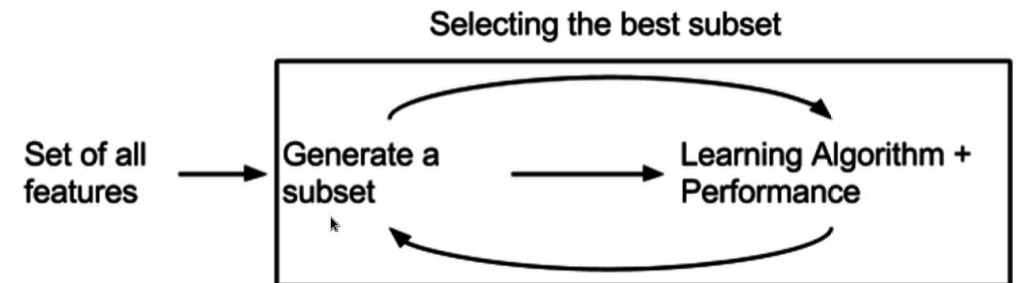
FILTER



WRAPPER



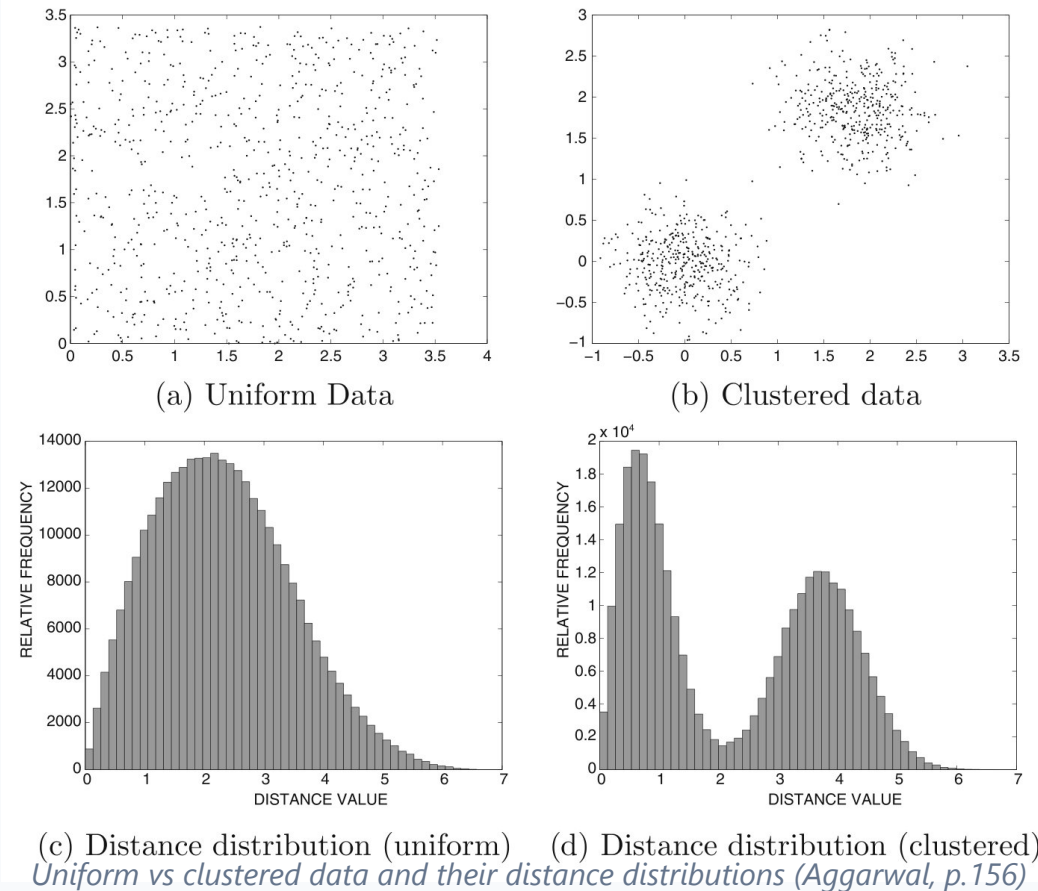
EMBEDDED



Filter, wrapper and embedded selection models

Without labels, the score must measure clustering tendency instead.

- Uniform data: one bell-shaped distance distribution. Clustered data: two peaks.
- Pick the subset that minimises the distance-distribution entropy.
- Named measures: term strength, classification-based relevance, Hopkins statistic.



Sometimes the right feature does not exist yet — so build it.

"In some cases, we may want to create new attributes based on others. Such attribute construction can help improve accuracy and understanding of structure in high-dimensional data." — Han, Kamber & Pei, p.105 "

- Example: $\text{area} = \text{height} \times \text{width}$ — a new column on the same axes.
- Selection removes; construction adds a column; extraction changes the axes.

Five traps decide this topic in the exam.

- Selection chooses existing; extraction and construction create.
- Gini and entropy: lower is better. Fisher score: higher is better.
- Filter is model-free; wrapper is model-tied.
- Fisher score selects a feature; Fisher's discriminant builds a direction.
- Hopkins near 1 means clustered; 0.5 means uniform.

The whole topic in five lines.

- Too many features hurt — noise and redundancy.
- We want the smallest subset that preserves class information.
- There are 2^n subsets, so we score and search greedily.
- The score is a filter, or model-tied (wrapper or embedded).
- Selection chooses; construction and extraction create.

References and image credits

- Han, Jiawei; Kamber, Micheline; Pei, Jian. Data Mining: Concepts and Techniques, 3rd ed. Morgan Kaufmann, 2011 — Section 3.4.4 (pp. 103-105).
- Aggarwal, Charu C. Data Mining: The Textbook. Springer, 2015 — Section 10.2 (pp. 287-293); Section 6.2 (pp. 155-158).
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